The Analysis of Multigroup Randomized Experiments in Large Sample Setups

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Abstract

Randomized experiments involving more than two treatment groups are central to many problems in causal inference. While existing work in experimental design has predominantly emphasized parametric and nonparametric methods suited to small-sample settings, large-sample methods remain comparatively underexplored. This paper presents a systematic introduction to the analysis of multigroup randomized experiments from a large-sample perspective. We discuss techniques for estimation, hypothesis testing — including simultaneous, interval, and equivalence hypotheses — confidence interval construction, power analysis, and sample size determination, for linear contrasts of the group parameters and for smooth functions of them, based on group sample means. The framework is illustrated through contrast constructions for common tasks: all-pairwise comparisons for identifying optimal treatments among G groups, many-to-one comparisons, tests of equality across groups, and full and fractional factorial designs.

1 Introduction

Under proper conditions randomized controlled trials (RCTs) are often cited as the gold standard for causal inference (Rubin 2008). One key goal is to estimate parameters that capture the causal influence of treatments on a selected population. Many specialized experimental designs are available. Some perform well for some tasks but may not be ideal for others.

For example, two-group RCTs invest the available sample size N into the comparison of two conditions. This makes it possible to maximize power for certain hypothesis tests under various allocation restrictions, like treatment costs or group size limitations (Jan and Shieh 2011). Multigroup[1] experiments with G

[1]Synonyms are multi-armed experiments, multivariable tests (MVTs), multi-armed trials or A/B/n tests.

groups allow the analysis of more — and different — questions in one experiment. For example, pairwise comparisons amongst G conditions — or a subset of them — make it possible to find the best-performing treatment(s) in a single run (Herberich et al. 2010). To screen efficiently through some or all of the G treatments, a joint hypothesis test can be carried out to test whether the expected values of the outcome are identical across the G groups, or a subset thereof. $G-1$ treatments can be compared against a baseline condition to screen for a better replacement treatment (Bretz et al. 2010). Testing G different dosages of a single factor allows the derivation of a dose-response relationship between the dosage and metrics of interest (Ritz et al. 2019). Testing combinations of dosages of several treatment factors allows the derivation of the response surface of the variable(s) of interest (Myers et al. 2016). In factorial experiments, groups are formed by distinct treatment factors in clever combinations simultaneously in a single experiment. Interaction between factors can be detected and can distinguish relevant from irrelevant factors via screening much faster than testing *one factor at a time* (OFAT), that is sequentially running two-group RCTs (Ledolter and Swersey 2007; Collins et al. 2014; Box et al. 1978; Wu and Hamada 2011; Hinkelmann and Kempthorne 2007, 2005; Hinkelmann 2012; Cox and Reid 2000).

Popular introductory *Design of Experiments* (DoE) textbooks (Montgomery 2017; Box et al. 1978; Wu and Hamada 2011; Dean and Voss 1999) mainly focus on applications for engineering, biomedical sciences, psychology and other social sciences. Small sample sizes are typical in these domains, so that the presented methods are tailored for that case. Usually, parametric methods are presented that are specialized for certain setups, defined by the outcome variable[2], the data type (i.e. binary, normal or count data), the sample size configuration (i.e. equal or unequal sample sizes across the G groups), and the group variances (i.e. homoscedasticity or heteroscedasticity). The parametric analysis of multigroup RCTs usually requires extensive manual statistical modeling, since different data types and sample size setups require specialized methods (Lewis et al. 1999). Additionally, assumption checking like testing normality of the outcome and variance homoscedasticity of the G groups is frequently required. Data transformation to make the data fit modeling assumptions is sometimes used, but can have unintended side effects, like changing the underlying null hypothesis (Feng et al. 2014). Nonparametric alternatives to parametric tests are available (Brunner and Puri 2001; Hettmansperger and McKean 2010). However, they may test different hypotheses than the parametric "counterpart" (Konietschke and Brunner 2023). Overall, multigroup RCT analysis in small samples is labor intensive and requires strong statistical expertise.

Most of these challenges disappear in large enough samples: Consistent and asymptotically normal estimators justify Wald-type tests and can be used for all numeric[3] data types (i.e. continuous, binary and count data), regardless

[2]Often the terms variable of interest, dependent variable or endpoint are interchangeably used in the statistical literature for the outcome variable.

[3]Categorical and ordinal outcomes need to be dummy encoded before this procedure can be

of designed unequal sample size distributions for the groups and regardless of heteroscedasticity. Extensive assumption checking and statistical modeling is not required. This makes automated analysis of experiments possible.

This large-sample approach to analyze multigroup RCTs is known (Pauly et al. 2015; Hothorn et al. 2008; Herberich et al. 2010), but receives considerably less attention in introductory DoE textbooks. The hypothesized reason is that large sample sizes are rare in domains for which the usual DoE textbooks are written — i.e. engineering, biomedical sciences, psychology and other social sciences. However, RCTs in business settings often have sample sizes in the hundreds of thousands (Lewis and Rao 2015; Kohavi et al. 2013), so that large-sample based methods for multigroup experiments are applicable.

The goal of this paper is to give an introduction to large-sample estimation and inference for multigroup randomized experiments. Techniques for different tasks are discussed, for example full and fractional factorial designs, multigroup tests with the aim of many-to-one and all-pairwise comparisons, and the reporting of effects in percent of a baseline.

Section 2 develops the general framework for estimation and global and simultaneous inference for linear and smooth transformations of the parameters; the parameters of interest in multigroup experiments — contrasts of group parameters — are defined and analyzed in Section 2.4. Section 3 develops the group-sample-means estimator and shows that it satisfies the framework's conditions. Section 4 constructs contrast matrices for common multigroup tasks.

2 Estimation and Inference in Large Sample Setups

Hothorn et al. (2008), Sec. 2, outlines a general framework for developing estimation and inference procedures for asymptotically normal estimators in large sample setups, whose variance-covariance matrix can be consistently estimated. The framework, for linear and smooth transformations of the parameters, is described in Section 2.1; simultaneous inference procedures are developed in Section 2.2, the global Wald test with power and sample size planning in Section 2.3; and the framework is applied to contrasts in Section 2.4.

2.1 General Framework

Following Hothorn et al. (2008), let $\mathbb{M}((S_1, \ldots, S_N), \theta, \eta)$ denote a semiparametric statistical model for a set of N observations $(S_1, \ldots, S_N)$. The model contains a fixed but unknown vector of elemental parameters of interest, $\theta := (\theta_1, \ldots, \theta_P) \in \mathbb{R}^P$, and possibly further (random or nuisance) parameters η.

applied for them.

The observations are deliberately generic: an observation S_i may be a cross-sectional scalar outcome or outcome–assignment–covariate record (Y_i, T_i, X_i), a multivariate endpoint, or a panel record of repeated measurements on one unit with arbitrary within-unit dependence and possibly unit-specific length, accommodating staggered entry, dropout, and irregular measurement times. No structure is imposed on the S_i beyond the two conditions Equation 1 and Equation 2 below. The setting of this paper is a randomized experiment with G treatment groups (Section 2.4.1), although the framework applies equally to nonrandomized studies and other estimation problems. P need not equal G, since a group may contribute several parameters — one per metric or subgroup — or none.

Denote θ's estimator as $\hat{\theta} := (\hat{\theta}_1, ..., \hat{\theta}_P)$ and its asymptotic variance-covariance matrix as $\Sigma^{\infty}_{\sqrt{N}\hat{\theta}}$, and write $\Sigma_{\hat{\theta}} := \Sigma^{\infty}_{\sqrt{N}\hat{\theta}}/N$ for the covariance matrix of the normal approximation at sample size N; the estimators are $\hat{\Sigma}^{\infty}_{\sqrt{N}\hat{\theta}}$ and $\hat{\Sigma}_{\hat{\theta}}$. The matrix $\Sigma_{\hat{\theta}}$ need not equal the finite-sample covariance of $\hat{\theta}$. Consider an asymptotically normal estimator $\hat{\theta}$,

$$\sqrt{N}(\hat{\theta} - \theta) \xrightarrow{D} N(0, \Sigma^{\infty}_{\sqrt{N}\hat{\theta}}) \tag{1}$$

whose variance-covariance matrix can be consistently estimated, in the sense

$$N \times \hat{\Sigma}_{\hat{\theta}} \xrightarrow{P} \Sigma^{\infty}_{\sqrt{N}\hat{\theta}}. \tag{2}$$

Accordingly, $\hat{\Sigma}^{\infty}_{\sqrt{N}\hat{\theta}} := N\hat{\Sigma}_{\hat{\theta}}$ serves as the estimator of $\Sigma^{\infty}_{\sqrt{N}\hat{\theta}}$. For large enough sample sizes $\hat{\theta}$ can be treated as approximately multivariate normal,

$$\hat{\theta} \overset{a}{\sim} N(\theta, \Sigma_{\hat{\theta}}),$$

with $\hat{\Sigma}_{\hat{\theta}}$ used as computable replacement for the unknown $\Sigma_{\hat{\theta}}$.

Consider linear transformations $\vartheta := H\theta$ of the parameter vector, specified through a constant matrix $H \in \mathbb{R}^{W \times P}$. The estimator $\hat{\vartheta} := H\hat{\theta}$ consistently estimates ϑ, is asymptotically normal, and can be treated as approximately normal in large samples,

$$\hat{\vartheta} \overset{a}{\sim} N(\vartheta, \Sigma_{\hat{\vartheta}}), \qquad \Sigma_{\hat{\vartheta}} = H\Sigma_{\hat{\theta}}H',$$

with the estimator $\hat{\Sigma}_{\hat{\vartheta}} = H\hat{\Sigma}_{\hat{\theta}}H'$ as computable replacement of the unknown $\Sigma_{\hat{\vartheta}}$.

The same holds for a smooth transformation $\vartheta := h(\theta)$, with its Jacobian in the role of H. Let h be differentiable in a neighbourhood of θ with a Jacobian that is continuous at θ,

$$\mathbf{J}(\theta) := \frac{\partial h(\theta)}{\partial \theta'} \in \mathbb{R}^{W \times P}, \qquad \mathbf{J}(\theta + u) \to \mathbf{J}(\theta) \quad (u \to 0). \tag{3}$$

Then $\hat{\vartheta} := h(\hat{\theta})$ consistently estimates ϑ. It is asymptotically normal (Serfling 1980, Thm. 3.3A) and can be treated as approximately normal in large samples,

$$\hat{\vartheta} \overset{a}{\sim} N(\vartheta, \Sigma_{\hat{\vartheta}}), \qquad \Sigma_{\hat{\vartheta}} = \mathbf{J}(\theta)\, \Sigma_{\hat{\theta}}\, \mathbf{J}(\theta)', \tag{4}$$

with the estimator $\hat{\Sigma}_{\hat{\vartheta}} = \mathbf{J}(\hat{\theta})\, \hat{\Sigma}_{\hat{\theta}}\, \mathbf{J}(\hat{\theta})'$ as computable replacement of the unknown $\Sigma_{\hat{\vartheta}}$, consistent in the sense of Equation 2 since $\mathbf{J}(\hat{\theta}) \xrightarrow{P} \mathbf{J}(\theta)$ by Equation 3.

The standardized estimator $Z := \hat{\Lambda}^{-1/2}(\hat{\vartheta} - \vartheta)$, with $\hat{\Lambda} := \operatorname{diag}(\hat{\Sigma}_{\hat{\vartheta}})$, is asymptotically normal and can be treated as approximately normal in large samples,

$$Z \overset{a}{\sim} N(0, \Sigma_Z), \qquad \Sigma_Z := \Lambda^{-1/2} \Sigma_{\hat{\vartheta}} \Lambda^{-1/2}, \quad \Lambda := \operatorname{diag}(\Sigma_{\hat{\vartheta}}),$$

with the estimator $\hat{\Sigma}_Z := \hat{\Lambda}^{-1/2} \hat{\Sigma}_{\hat{\vartheta}} \hat{\Lambda}^{-1/2}$ as computable replacement of the unknown Σ_Z, the asymptotic correlation matrix of $\hat{\vartheta}$; both have unit diagonal.

The statements of this section are large-sample approximations: at finite N the distribution of $\hat{\theta}$ need not be normal, $\hat{\Sigma}_{\hat{\theta}}$ need not match the finite-sample covariance, and the actual level of a test can differ from the nominal one (Kiviet 2012, sec. 4). By how much is a question for Monte Carlo simulation (Kiviet 2012, secs. 2–5): fix a data generating process within $\mathbb{M}$ at a given sample size, draw independent samples $r = 1, 2, \ldots$ from it, and compute in each the estimates $\hat{\theta}^{(r)}$ and $\hat{\Sigma}^{(r)}_{\hat{\theta}}$. Their empirical distribution estimates the sampling distribution of $\hat{\theta}$ and $\hat{\Sigma}_{\hat{\theta}}$, and hence the bias and variance of $\hat{\theta}$, the distribution of Z, and the actual rejection and coverage probabilities; a grid of sample sizes, parameter values and distributions shows where the approximations hold.

2.2 Simultaneous Inference

We test W null hypotheses on the coordinates of ϑ,

$$H_0^w \colon \vartheta_w = \vartheta_{0w}, \qquad w = 1, \ldots, W,$$

with the hypothesized values collected in $\vartheta_0 := (\vartheta_{01}, \ldots, \vartheta_{0W})'$. Jointly they form the hypothesis

$$H_0 \colon \ \vartheta = \vartheta_0 \text{ vs. } H_1 \colon \ \vartheta \neq \vartheta_0,$$

the *general linear hypothesis* $H\theta = \vartheta_0$ of Hothorn et al. (2008) when $\vartheta = H\theta$; for $\vartheta = h(\theta)$ it is $I_W \vartheta = \vartheta_0$, the identity matrix taking the role of H. Under H_0 we have

$$Z = \hat{\Lambda}^{-1/2}(\hat{\vartheta} - \vartheta_0) \overset{a}{\sim} N(0, \Sigma_Z),$$

with $\hat{\Sigma}_Z$ as computable replacement of the unknown Σ_Z. From here on Z denotes this computable test statistic, centered at ϑ_0.

Testing $H_0^1, \ldots, H_0^W$ together, as a *family* of hypotheses, requires an error criterion for the family rather than for each test alone, and a procedure that controls it. The criterion of this paper is the *familywise error rate* (FWER), the probability

of rejecting at least one true hypothesis (Hothorn et al. 2008; Bretz et al. 2010); a procedure controls it at level α if this probability is asymptotically at most α, whichever of the H_0^w are true.

Single-step procedures reject H_0^w whenever $|Z_w| > q$ for a common critical value q, chosen such that under H_0, where the FWER is largest,

$$\text{FWER} = P\Big[\max_w |Z_w| > q\Big] \le \alpha$$

in large samples.

The *Bonferroni* procedure uses $q = z_{1-\alpha/(2W)}$, where z_p denotes the p-quantile of the standard normal distribution with cdf $\Phi_{0,1}$. By the union bound,

$$P\Big[\max_w |Z_w| > q\Big] \;\le\; \sum_{w=1}^{W} P\big[|Z_w| > q\big] \;\to\; \alpha,$$

since each term tends to α/W; the FWER is thus asymptotically at most α, whatever the correlations in Σ_Z.

The *Šidák* procedure uses $q = z_{1-\alpha_S/2}$ with $\alpha_S := 1-(1-\alpha)^{1/W}$. If the coordinates of Z were independent, the FWER would tend to exactly α,

$$P\Big[\max_w |Z_w| \le q\Big] \;\to\; (1-\alpha_S)^W = 1-\alpha;$$

for any correlation matrix the probability is at least as large (Šidák 1967), so the FWER is asymptotically at most α. Since $\alpha_S > \alpha/W$, the critical value is slightly smaller than Bonferroni's.

The *single-step max-t procedure* (Hothorn et al. 2008) uses the $(1-\alpha)$-quantile of $\max_w |\zeta_w|$ for $\zeta \sim N(0, \hat\Sigma_Z)$, $u_{1-\alpha} := u_{1-\alpha}(\hat\Sigma_Z)$ defined by

$$P\Big[\max_w |\zeta_w| \le u_{1-\alpha}(\hat\Sigma_Z)\Big] = 1-\alpha$$

and computed by numerical integration (Genz and Bretz 2009); with $q = u_{1-\alpha}$ the FWER tends to α.

Together with the per-test value $z_{1-\alpha/2}$, the three critical values are ordered,

$$z_{1-\alpha/2} \;\le\; u_{1-\alpha}(\hat\Sigma_Z) \;\le\; z_{1-\alpha_S/2} \;\le\; z_{1-\alpha/(2W)},$$

with equality in the middle when $\hat\Sigma_Z = I_W$ and on the left when all correlations are ± 1: the max-t procedure is the only one that uses the estimated correlations, and it is never more conservative than the other two.

Replacing ϑ_0 by ϑ turns Z into the standardized estimator of Section 2.1, which is asymptotically $N(0, \Sigma_Z)$ under any hypothesis. For each of the three critical values q the W intervals

$$\hat\vartheta_w \pm q\sqrt{(\hat\Sigma_{\hat\vartheta})_{ww}}, \qquad w = 1, \dots, W,$$

are therefore *simultaneous confidence intervals*: they cover all their parameters with asymptotic probability $P[\max_w |\zeta_w| \leq q]$, at least $1-\alpha$, and exactly $1-\alpha$ with the shortest intervals for $q = u_{1-\alpha}$.

2.3 Global Test

Beyond deciding on the W hypotheses H_0^w one at a time with familywise calibration (Section 2.2), H_0 can be tested as one joint hypothesis with the well-known Wald test. If $\Sigma_{\hat{\vartheta}}$ is positive definite, its test statistic, the *Wald-type statistic* (WTS), is asymptotically χ^2 distributed with W degrees of freedom under H_0,

$$\begin{aligned} WTS &:= (\hat{\vartheta} - \vartheta_0)' \hat{\Sigma}_{\hat{\vartheta}}^{-1} (\hat{\vartheta} - \vartheta_0) \\ &= N(\hat{\vartheta} - \vartheta_0)' (H \hat{\Sigma}^{\infty}_{\sqrt{N}\hat{\theta}} H')^{-1} (\hat{\vartheta} - \vartheta_0) \xrightarrow[N\to\infty]{D} \chi^2_W. \end{aligned} \tag{5}$$

Under local alternatives, with $\vartheta - \vartheta_0$ of order $N^{-1/2}$, the WTS approximately follows a noncentral χ^2 distribution (Rao 1973, sec. 3b),

$$\begin{aligned} WTS &\overset{a}{\sim} \chi^2_W(\lambda_N), \qquad \lambda_N = N\Delta^2, \\ \Delta^2 &= (\vartheta - \vartheta_0)' \left(H \Sigma^{\infty}_{\sqrt{N}\hat{\theta}} H' \right)^{-1} (\vartheta - \vartheta_0). \end{aligned} \tag{6}$$

This approximation is standard large-sample theory (Stroud 1973; Vaart 1998, Ch. 16) and is used for planning at fixed alternatives as well (accuracy for $W = 1$, where $WTS = Z^2$: Appendix, derivation of Equation 11). The level-α test rejects the null hypothesis if $WTS > \chi^2_{W,1-\alpha}$. Its approximate power is

$$\pi(\theta, N) = 1 - F_{W,\lambda_N}\left(\chi^2_{W,1-\alpha}\right), \tag{7}$$

where $F_{W,\xi}$ denotes the cdf of $\chi^2_W(\xi)$, which is strictly decreasing in ξ (Ghosh 1973). For a target power $1-\beta > \alpha$ the minimal sample size is therefore $N^* = \lceil \xi^*(W, \alpha, \beta)/\Delta^2 \rceil$, with ξ^* the unique root of $F_{W,\xi}(\chi^2_{W,1-\alpha}) = \beta$ (Shieh 2005). The noncentrality input Δ^2 is estimator-specific; for group sample means it becomes explicit in Section 4.4.

In small samples the χ^2_W calibration can be liberal; power is then better assessed by simulation (Pauly et al. 2015; Konietschke et al. 2015).

2.4 Application to Contrasts

2.4.1 Contrasts in Multigroup Experiments

Consider a randomized experiment with G groups. Each group g receives a different treatment, which may arise either from a single factor[4] f with $L_f = G$ levels, or from a combination of K distinct factors $f_1, \ldots, f_K$, each with L_{f_k}

[4] A frequently used synonym in the experimentation domain is independent variable (IV).

levels. Let the distribution of the outcome metric Y under group g be denoted F_g, and write $Y(g)$ for the (potential) outcome of Y under assignment to group g, so that $Y(g) \sim F_g$. The primary interest is typically in comparing parameters θ_g of F_g, such as means or medians. A principled way to compare parameters across conditions is via contrasts. A contrast C is defined as a linear combination of the parameters $\theta = (\theta_1, \dots, \theta_G)'$ — here the framework's dimension is $P = G$, one parameter per group — with fixed, user-specified weights $c = (c_1, \dots, c_G)$ satisfying $\sum_{g=1}^{G} c_g = 0$,

$$C := c\theta = \sum_{g=1}^{G} c_g \theta_g.$$

A comprehensive introduction to contrasts is provided by Wiens and Nilsson (2017). A contrast involving only two parameters is known as a *pairwise comparison*. For instance, the average treatment effect (ATE) comparing group 1 to group 2 is obtained, when $\theta_g = \mu_g := E[Y(g)]$ is the group mean, using the weight vector $c = (1, -1, 0, \dots, 0)$, resulting in

$$C = c\theta = \theta_1 - \theta_2 = \mu_1 - \mu_2 =: \text{ATE}(1, 2).$$

As a second example, consider a 2×2 full factorial experiment with two binary factors, denoted A and B. Each factor has two levels, labeled generically as level 0 and level 1, so that the four treatment groups correspond to the combinations $(A = 0, B = 0)$, $(A = 1, B = 0)$, $(A = 0, B = 1)$, and $(A = 1, B = 1)$. Let $\mu_1, \mu_2, \mu_3, \mu_4$ denote the corresponding group means. The main effect of factor A can be expressed as the average difference in outcomes when A changes from level 0 to level 1, averaged over the levels of B,

$$C = \frac{1}{2}(\mu_2 + \mu_4) - \frac{1}{2}(\mu_1 + \mu_3),$$

with associated contrast weights $c = (-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$. The interaction effect between A and B corresponds to the extent to which the effect of A differs across levels of B,

$$C = \mu_1 - \mu_2 - \mu_3 + \mu_4,$$

with contrast weights $c = (1, -1, -1, 1)$. Multiple contrasts can be evaluated jointly by stacking the corresponding weight vectors $c^{(w)} = (c_{w1}, \dots, c_{wG})$ for $w = 1, \dots, W$ into a $W \times G$ contrast matrix H, yielding a vector of contrasts, $C = H\theta$.

2.4.2 Tests and Confidence Intervals

The framework applies to contrasts by setting $H = c = (c_1, ..., c_G)$. The contrast and its estimator are given by

$$\vartheta = H\theta = \sum_{g=1}^{G} c_g \theta_g =: C \text{ and } \hat{\vartheta} = H\hat{\theta} = \sum_{g=1}^{G} c_g \hat{\theta}_g =: \hat{C}.$$

For large enough sample sizes it follows that $\hat{\vartheta} = \hat{C} \overset{a}{\sim} N(C, \Sigma_{\hat{C}})$, with

$$\Sigma_{\hat{C}} = c\Sigma_{\hat{\theta}}c' = \sum_{g=1}^{G} c_g^2(\Sigma_{\hat{\theta}})_{gg} + 2\sum_{g=1}^{G}\sum_{l:l>g}^{G} c_g c_l(\Sigma_{\hat{\theta}})_{gl} \tag{8}$$

which can be estimated using

$$\widehat{\mathrm{Var}}[\hat{C}] = \hat{\Sigma}_{\hat{C}} = c\hat{\Sigma}_{\hat{\theta}}c' = \sum_{g=1}^{G} c_g^2\widehat{\mathrm{Var}}[\hat{\theta}_g] + 2\sum_{g=1}^{G}\sum_{l:l>g}^{G} c_g c_l\widehat{\mathrm{Cov}}[\hat{\theta}_g, \hat{\theta}_l]. \tag{9}$$

If $\hat{\theta}$ is a consistent estimator for θ, $\hat{C}$ is a consistent estimator for a contrast C. The null hypothesis for a two-tailed test is

$$H_0 \colon\ C = C_0 \text{ vs. } H_1 \colon\ C \neq C_0,$$

and the test statistic Z of Section 2.2, here scalar, is asymptotically standard normal under H_0,

$$Z = \hat{\Sigma}_{\hat{C}}^{-1/2}(\hat{C} - C_0) = \frac{\hat{C} - C_0}{\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}} \xrightarrow[N\to\infty]{D} N(0,1). \tag{10}$$

Inverting it over C_0 gives the $1-\alpha$ confidence set

$$\mathcal{C}_{two}^{1-\alpha} := [\hat{C} - z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}[\hat{C}]},\ \hat{C} + z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}].$$

The null hypothesis is rejected at level α if the realized statistic $Z = z$ falls into the rejection region, if the p-value — the tail probability beyond z under H_0 — is below α, or if C_0 lies outside the confidence set; the three are one criterion:

$$\begin{aligned} & z \in \mathcal{R}_{two} = \{z : |z| > z_{1-\alpha/2}\} \\ & \iff p_{two} = 2\left(1 - \Phi_{0,1}(|z|)\right) < \alpha \\ & \iff C_0 \notin \mathcal{C}_{two}^{1-\alpha}. \end{aligned}$$

For a right-tailed test the null hypothesis

$$H_0 \colon\ C \leq C_0 \text{ vs. } H_1 \colon\ C > C_0$$

has the confidence set

$$\mathcal{C}_{right}^{1-\alpha} := [\hat{C} - z_{1-\alpha}\sqrt{\widehat{\mathrm{Var}}[\hat{C}]},\ \infty)$$

and is rejected at level α if

$$\begin{aligned} & z \in \mathcal{R}_{right} = \{z : z > z_{1-\alpha}\} \\ & \iff p_{right} = 1 - \Phi_{0,1}(z) < \alpha \\ & \iff C_0 \notin \mathcal{C}_{right}^{1-\alpha}. \end{aligned}$$

For a left-tailed test the null hypothesis

$$H_0 \colon\ C \geq C_0 \text{ vs. } H_1 \colon\ C < C_0$$

has the confidence set

$$\mathcal{C}_{left}^{1-\alpha} := (-\infty,\ \hat{C} + z_{1-\alpha}\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}]$$

and is rejected at level α if

$$\begin{aligned} & z \in \mathcal{R}_{left} = \{z : z < -z_{1-\alpha}\} \\ & \iff p_{left} = \Phi_{0,1}(z) < \alpha \\ & \iff C_0 \notin \mathcal{C}_{left}^{1-\alpha}. \end{aligned}$$

Point null hypotheses of the form $H_0 : C = C_0$ are often unrealistically precise and rarely, if ever, true. Additionally, even tiny differences from C_0 can lead to rejection — here "statistical significance" might not lead to "practical significance". A general remedy is *interval null hypothesis testing*, which replaces the point C_0 by a pre-specified interval $[C_L, C_U]$ of practically irrelevant values and works with the two-sided confidence set at level $1-2\alpha$, $\mathcal{C}_{two}^{1-2\alpha}$, and the one-sided statistics

$$Z_L := \frac{\hat{C} - C_L}{\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}}, \qquad Z_U := \frac{C_U - \hat{C}}{\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}}.$$

The hypothesis

$$H_0 \colon\ C \in [C_L, C_U] \text{ vs. } H_1 \colon\ C \notin [C_L, C_U]$$

is rejected at level α if

$$\begin{aligned} & \min\{z_L, z_U\} < -z_{1-\alpha} \\ & \iff p = \min\{\Phi_{0,1}(z_L),\ \Phi_{0,1}(z_U)\} < \alpha \\ & \iff \mathcal{C}_{two}^{1-2\alpha} \cap [C_L, C_U] = \emptyset, \end{aligned}$$

and the *equivalence* hypothesis

$$H_0 \colon\ C \notin [C_L, C_U] \text{ vs. } H_1 \colon\ C \in [C_L, C_U],$$

tested by two one-sided tests (TOST) (Schuirmann 1987; Wellek 2010), is rejected at level α if

$$\begin{aligned} & \min\{z_L, z_U\} > z_{1-\alpha} \\ & \iff p = \max\{1 - \Phi_{0,1}(z_L),\ 1 - \Phi_{0,1}(z_U)\} < \alpha \\ & \iff \mathcal{C}_{two}^{1-2\alpha} \subseteq [C_L, C_U]. \end{aligned}$$

Both tests have asymptotic size α, attained on the boundary of the null interval. The *minimum effect test* (superiority test) is the interval null test with $[C_L, C_U] =$

$[-\delta_0, \delta_0]$:[5] an effect is declared practically relevant when $\mathcal{C}_{two}^{1-2\alpha}$ lies entirely outside $[-\delta_0, \delta_0]$.

For families of contrasts, stacked into a contrast matrix H, the simultaneous and global procedures of Section 2.2 and Section 2.3 apply directly: in the two-sided rules and confidence sets above, $z_{1-\alpha/2}$ is replaced by the common critical value q of Section 2.2, which turns $\mathcal{C}_{two}^{1-\alpha}$, row by row, into the simultaneous confidence intervals of that section.

2.4.3 Power Analysis and Sample Size Planning

The *effect* $\delta := C - C_0$ measures the displacement of the true contrast value C from the hypothesized value C_0; the alternative corresponds to $\delta \neq 0$. Under local alternatives, with δ of order $N^{-1/2}$, the test statistic Equation 10 is approximately noncentral normal (Vaart 1998) (derivation and scope in the Appendix),

$$\begin{aligned} Z &\overset{a}{\sim} N(\lambda, 1), \\ \lambda &:= \delta\sqrt{N/\bar{V}_C}, \\ \bar{V}_C &:= c\,\Sigma_{\sqrt{N}\hat{\theta}}^{\infty}\,c'. \end{aligned} \tag{11}$$

The power of the two-sided level-α test is, up to a conservative approximation (Appendix),

$$\pi \approx \Phi_{0,1}(|\lambda| - z_{1-\alpha/2}). \tag{12}$$

Setting the target $\pi = 1 - \beta$ and inverting yields one equation in the five quantities $(\alpha, \beta, \delta, \bar{V}_C, N)$; a power calculation fixes four of them and solves for the fifth (Cohen 1988). Solving for the sample size and, at fixed N, for the *minimum detectable effect* gives

$$N \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2\,\bar{V}_C}{\delta^2}, \qquad \delta_{\min} = (z_{1-\alpha/2} + z_{1-\beta})\sqrt{\bar{V}_C/N}; \tag{13}$$

for one-sided tests $z_{1-\alpha/2}$ is replaced by $z_{1-\alpha}$. Power depends on the estimator and design only through $\bar{V}_C$; its explicit form, and its optimization over the design, are given for group sample means in Section 4.1.

Sample size planning can also target the length $L := 2\,z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}$ of the confidence interval the experiment will report — *accuracy in parameter estimation* (AIPE) or planning for precision (Kelley et al. 2003; Maxwell et al. 2008; Rothman and Greenland 2018; Lai and Kelley 2012; Liu 2009; Grieve 1991) —

[5] Not to be confused with the effect $\delta = C - C_0$ of Section 2.4.3.

which requires no postulated effect size. Since $N\,\widehat{\mathrm{Var}}[\hat{C}] \xrightarrow{P} \bar{V}_C$, the expected length and the sample size reaching a target expected length ℓ are

$$E[L] \approx 2\, z_{1-\alpha/2}\sqrt{\bar{V}_C/N}, \qquad N \geq \frac{4\, z_{1-\alpha/2}^2\, \bar{V}_C}{\ell^2}.$$

Conversely, an experiment powered at $(\alpha, 1-\beta)$ for an effect δ reports intervals of expected length

$$E[L] \approx 2z_{1-\alpha/2}\, \delta/(z_{1-\alpha/2} + z_{1-\beta}),$$

which is $\approx 1.4\,\delta$ at $(\alpha, 1-\beta) = (0.05, 0.80)$. Since L fluctuates around its expectation, a design meeting $E[L] \leq \ell$ still produces $L > \ell$ in roughly half of all realizations.

3 Estimators

The general framework in Section 2 requires only consistent and asymptotically normal estimators with consistently estimable asymptotic variance-covariance matrices.

3.1 Group Sample Means

Sample-mean based estimators in combination with the multivariate model of Konietschke et al. (2015) and Friedrich et al. (2017) fulfill these criteria, as shown below. The model imposes minimal restrictions on the data generating process: it assumes neither multivariate normality nor homogeneous covariance matrices, allows for different sample sizes, and permits the distributions of the error vectors to differ across groups. Consider an experiment with G treatment groups and a numeric outcome vector $Y_{ig} = (Y_{ig1}, ..., Y_{igM})'$ observed for unit i assigned to group g. Throughout, I_M denotes the $M \times M$ identity matrix, $\mathbb{1}_M$ and $\mathbb{0}_M$ the M-dimensional column vectors of ones and zeros, and $\mathbb{1}\{\cdot\}$ the indicator function. The assumptions are as follows:

$$Y_{ig} = \mu_g + \varepsilon_{ig}, \quad i = 1, ..., N_g, \quad g = 1, ..., G, \quad N = N_1 + \cdots + N_G, \tag{14}$$

$$\begin{aligned} E[\varepsilon_{ig}] &= \mathbb{0}_M, \\ \mathrm{Var}[\varepsilon_{ig}] &= \Sigma_g > 0, \\ E[\|\varepsilon_{ig}\|^4] &< \infty, \end{aligned} \tag{15}$$

$$N_g/N \xrightarrow{N\to\infty} p_g \in (0,1), \tag{16}$$

where the error vectors are independent across all units and identically distributed within each group. This is a nonparametric model: no distributional assumptions are imposed on the errors, which are only required to have finite variance-covariance matrices and finite fourth moments. No special structure of the covariance matrix Σ_g is assumed and the Σ_g may differ between groups; the number of coordinates may in general also differ between groups by design (Friedrich et al. 2017), but we take a common, fixed number M throughout. The individual vectors are aggregated into

$$\begin{aligned}\mathbf{Y} &= (Y_{11}', ..., Y_{N_G G}')', \\ \mu &= (\mu_1', ..., \mu_G')', \\ \mu_g &= (\mu_{g1}, ..., \mu_{gM})',\end{aligned}$$

and this general design allows for factorial structures within the components of μ by splitting up the indices.

In this section the elemental parameter of Section 2.1 is the stacked mean vector, $\theta = \mu$ — so that $P = GM$ — estimated by the group sample means. Denote the group sample mean vector and sample variance-covariance matrix as

$$\bar{\mu}_g := \frac{1}{N_g}\sum_{i=1}^{N_g} Y_{ig}, \quad \hat{\Sigma}_g := \frac{1}{N_g - 1}\sum_{i=1}^{N_g}(Y_{ig} - \bar{\mu}_g)(Y_{ig} - \bar{\mu}_g)'.$$

The estimator of μ is $\bar{\mu} = (\bar{\mu}_1', ..., \bar{\mu}_G')'$. The exact covariance matrix of the normalized vector of group means and its empirical estimator are the block-diagonal matrices

$$\begin{aligned}\Sigma_{\sqrt{N}\bar{\mu}} &= \mathrm{Var}[\sqrt{N}\bar{\mu}] = \mathrm{diag}\left\{\frac{N}{N_g}\Sigma_g : 1 \le g \le G\right\}, \\ \hat{\Sigma}_{\sqrt{N}\bar{\mu}} &= \mathrm{diag}\left\{\frac{N}{N_g}\hat{\Sigma}_g : 1 \le g \le G\right\};\end{aligned} \tag{17}$$

equivalently, $\mathrm{Var}[\bar{\mu}]$ is estimated by

$$\hat{\Sigma}_{\bar{\mu}} := \hat{\Sigma}_{\sqrt{N}\bar{\mu}}/N = \mathrm{diag}\{\hat{\Sigma}_g/N_g : 1 \le g \le G\},$$

which plays the role of the estimator $\hat{\Sigma}_{\hat{\theta}}$ of Section 2.1.

By the multivariate central limit theorem (Konietschke et al. 2015, Thm. 2.1), $\bar{\mu}$ is asymptotically normal,

$$\sqrt{N}(\bar{\mu} - \mu) \xrightarrow{D} N(0, \Sigma^{\infty}_{\sqrt{N}\bar{\mu}}), \qquad \Sigma^{\infty}_{\sqrt{N}\bar{\mu}} := \mathrm{diag}\{\Sigma_1/p_1, \dots, \Sigma_G/p_G\}, \tag{18}$$

and the estimator Equation 17 is consistent, $\hat{\Sigma}_{\sqrt{N}\bar{\mu}} \to \Sigma^{\infty}_{\sqrt{N}\bar{\mu}}$ almost surely , with 16. Group sample means thus satisfy Equation 1 and Equation 2.

The following examples instantiate Equation 14 and Equation 17 for four canonical designs covered by the model.

Example 1 (Univariate cross-sectional). *Each unit is measured on a single metric at a single occasion.*

This case, the workhorse of single-metric experiments, has $M = 1$: the observation vector is the scalar Y_{ig}, the mean vector is the scalar $\mu_g = E[Y_{ig}]$, so that $\theta = \mu = (\mu_1, \dots, \mu_G)'$, and the covariance matrix in Equation 15 reduces to the variance $\sigma_g^2 := \mathrm{Var}[Y_{ig}]$. The group sample mean estimator becomes

$$\bar{\mu}_g = \frac{1}{N_g} \sum_{i=1}^{N_g} Y_{ig},$$

and the estimator of its sampling variance σ_g^2/N_g becomes $\hat{\sigma}_g^2/N_g$ with

$$\hat{\sigma}_g^2 := \frac{1}{N_g - 1} \sum_{i=1}^{N_g} (Y_{ig} - \bar{\mu}_g)^2.$$

The matrices Equation 17 become diagonal,

$$\Sigma_{\sqrt{N}\bar{\mu}} = \begin{pmatrix} \frac{N}{N_1}\sigma_1^2 & 0 & \cdots & 0 \\ 0 & \frac{N}{N_2}\sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{N}{N_G}\sigma_G^2 \end{pmatrix},$$

of dimension $G \times G$, and $\hat{\Sigma}_{\sqrt{N}\bar{\mu}}$ has the same form with $\hat{\sigma}_g^2$ in place of σ_g^2. This recovers the univariate model of Pauly et al. (2015).

Estimation requires only the group summaries $\bar{\mu}_g$, N_g and $\hat{\sigma}_g^2$. Equivalently, stack observations by group and then by unit, and let $\mathbf{D}$ be the $N \times G$ matrix of group indicators, with entry $D_{(ig),g'} = \mathbb{1}\{g = g'\}$. The linear working model has no separate intercept:

$$\begin{aligned} \mathbf{Y} &= \mathbf{D}\mu + \varepsilon, \\ \Omega := \mathrm{Var}[\varepsilon] &= \mathrm{diag}\{\sigma_g^2 I_{N_g} : 1 \le g \le G\}. \end{aligned}$$

Its OLS coefficient vector equals $\bar{\mu}$ (Rencher and Schaalje 2008, Ch. 13), and for $N_g > 1$ the HC2 version of the Eicker–Huber–White sandwich covariance, built from the residuals $\hat{\varepsilon}_{ig} := Y_{ig} - \bar{\mu}_g$ as

$$\hat{\Omega}_{\mathrm{HC2}} := \mathrm{diag}\{\hat{\Omega}_1, \dots, \hat{\Omega}_G\}, \qquad \hat{\Omega}_g := \frac{N_g}{N_g - 1} \mathrm{diag}\{\hat{\varepsilon}_{1g}^2, \dots, \hat{\varepsilon}_{N_g g}^2\},$$

reproduces $\hat{\Sigma}_{\bar{\mu}}$ exactly:

$$\begin{aligned} \hat{\Sigma}_{\bar{\mu}}^{\mathrm{HC2}} &:= (\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'\hat{\Omega}_{\mathrm{HC2}}\mathbf{D}(\mathbf{D}'\mathbf{D})^{-1} \\ &= \mathrm{diag}\{N_g^{-1}\}\ \mathrm{diag}\{N_g\hat{\sigma}_g^2\}\ \mathrm{diag}\{N_g^{-1}\} \\ &= \mathrm{diag}\{\hat{\sigma}_g^2/N_g : 1 \le g \le G\} = \hat{\Sigma}_{\bar{\mu}}, \end{aligned} \tag{19}$$

because $\mathbf{D}'\mathbf{D} = \mathrm{diag}\{N_g\}$ and $\mathbf{D}'\hat{\Omega}_{\mathrm{HC2}}\mathbf{D} = \mathrm{diag}\{\frac{N_g}{N_g-1}\sum_{i=1}^{N_g}\hat{\varepsilon}_{ig}^2\} = \mathrm{diag}\{N_g\hat{\sigma}_g^2\}$. Multiplication by N gives $\hat{\Sigma}_{\sqrt{N}\bar{\mu}}$ in Equation 17.

Example 2 (Univariate panel). *Each unit is measured on a single metric at each of several fixed occasions, with treatment assigned once at the unit level and held fixed across occasions.*

Extend Example 1 by observing the same metric at T fixed occasions. Let Y_{igt} denote the value of the metric recorded on unit i of group g at occasion $t = 1, \dots, T$. The observation vector collects the unit's time course,

$$Y_{ig} = (Y_{ig1}, \dots, Y_{igT})',$$

so that Equation 14 holds with $M = T$ and, componentwise,

$$Y_{igt} = \mu_{gt} + \varepsilon_{igt}, \qquad \mu_{gt} = E[Y_{igt}],$$

which centers each observation at its group–time mean. Writing $\sigma_{g,tt'} := \mathrm{Cov}[Y_{igt}, Y_{igt'}]$, the covariance matrix in Equation 15 is

$$\Sigma_g = \begin{pmatrix} \sigma_{g,11} & \sigma_{g,12} & \cdots & \sigma_{g,1T} \\ \sigma_{g,12} & \sigma_{g,22} & \cdots & \sigma_{g,2T} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{g,1T} & \sigma_{g,2T} & \cdots & \sigma_{g,TT} \end{pmatrix},$$

carrying the variances of the metric at each occasion on its diagonal and its serial covariances off it.

The assumptions in Equation 15 concern the joint error vector across occasions; neither smoothness of the mean trajectory nor stationarity of the outcome process is required. The fixed population means μ_{gt} may therefore exhibit trends, seasonality, or abrupt level changes. With T fixed, each group–time mean is consistently estimated by averaging across independent units regardless of the shape of the mean trajectory. The covariance estimator likewise centers each occasion by its own sample mean, so changes in the population mean across occasions do not enter the estimated error covariance.

Temporal dependence within a unit is retained in Σ_g. Processes with random intercepts or slopes, serially correlated errors, or unit-specific random walks are covered whenever their error vectors are identically distributed within groups, independent across units, and satisfy the stated moment and positive-definiteness conditions. Common stochastic shocks that induce dependence between units fall outside this formulation.

The group sample mean estimator becomes the vector of occasion means,

$$\bar{\mu}_g = (\bar{\mu}_{g1}, \dots, \bar{\mu}_{gT})', \qquad \bar{\mu}_{gt} = \frac{1}{N_g}\sum_{i=1}^{N_g} Y_{igt},$$

and the estimator of its sampling variance Σ_g/N_g becomes $\hat{\Sigma}_g/N_g$ with entries

$$\hat{\sigma}_{g,tt'} := \frac{1}{N_g - 1} \sum_{i=1}^{N_g} (Y_{igt} - \bar{\mu}_{gt})(Y_{igt'} - \bar{\mu}_{gt'}).$$

The matrices Equation 17 are block-diagonal with G blocks of size $T \times T$, giving total dimension $GT \times GT$. This is the repeated measures setting of Friedrich et al. (2017).

Example 3 (Multivariate cross-sectional). *Each unit is measured on several metrics at a single occasion.*

Extend Example 1 by recording V metrics at a single occasion. Let Y_{igv} denote the value of metric $v = 1, \dots, V$ recorded on unit i of group g. The observation vector collects the unit's metrics,

$$Y_{ig} = (Y_{ig1}, ..., Y_{igV})',$$

so that Equation 14 holds with $M = V$ and $\mu_{gv} = E[Y_{igv}]$. Writing $\sigma_{g,vv'} := \mathrm{Cov}[Y_{igv}, Y_{igv'}]$, the covariance matrix in Equation 15 is

$$\Sigma_g = \begin{pmatrix} \sigma_{g,11} & \sigma_{g,12} & \cdots & \sigma_{g,1V} \\ \sigma_{g,12} & \sigma_{g,22} & \cdots & \sigma_{g,2V} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{g,1V} & \sigma_{g,2V} & \cdots & \sigma_{g,VV} \end{pmatrix},$$

carrying the variances of the metrics on its diagonal and the covariances between metrics on the same unit, contemporaneous by construction, off it. The group sample mean estimator becomes the vector of metric means,

$$\bar{\mu}_g = (\bar{\mu}_{g1}, \dots, \bar{\mu}_{gV})', \qquad \bar{\mu}_{gv} = \frac{1}{N_g} \sum_{i=1}^{N_g} Y_{igv},$$

and the estimator of its sampling variance Σ_g/N_g becomes $\hat{\Sigma}_g/N_g$ with entries

$$\hat{\sigma}_{g,vv'} := \frac{1}{N_g - 1} \sum_{i=1}^{N_g} (Y_{igv} - \bar{\mu}_{gv})(Y_{igv'} - \bar{\mu}_{gv'}).$$

The matrices Equation 17 are block-diagonal with G blocks of size $V \times V$, giving total dimension $GV \times GV$. This is the general MANOVA setting of Konietschke et al. (2015).

Example 4 (Multivariate panel). *Each unit is measured on several metrics at each of several fixed occasions, with treatment assigned once at the unit level and held fixed across occasions.*

Combining the repeated measurements of Example 2 with the multiple metrics of Example 3, let Y_{igvt} denote the value of metric $v = 1, \dots, V$ recorded on unit

i of group g at occasion $t = 1, \ldots, T$. Collect first the time course of a single metric v on unit i in the $T \times 1$ vector

$$Y_{ig}^{(v)} = (Y_{igv1}, ..., Y_{igvT})',$$

and then stack the V metric vectors into the unit's full observation vector,

$$Y_{ig} = (Y_{ig}^{(1)\prime}, ..., Y_{ig}^{(V)\prime})',$$

so that Equation 14 holds with $M = VT$ and $\mu_{gvt} = E[Y_{igvt}]$.

The covariance matrix in Equation 15 is built up along the same two steps. Writing $\sigma_{g,tt'}^{(vv')} := \mathrm{Cov}[Y_{igvt}, Y_{igv't'}]$, the covariance matrix of a single metric vector is the $T \times T$ matrix

$$\Sigma_g^{(vv)} := \mathrm{Var}[Y_{ig}^{(v)}] = \begin{pmatrix} \sigma_{g,11}^{(vv)} & \sigma_{g,12}^{(vv)} & \cdots & \sigma_{g,1T}^{(vv)} \\ \sigma_{g,12}^{(vv)} & \sigma_{g,22}^{(vv)} & \cdots & \sigma_{g,2T}^{(vv)} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{g,1T}^{(vv)} & \sigma_{g,2T}^{(vv)} & \cdots & \sigma_{g,TT}^{(vv)} \end{pmatrix},$$

carrying the variances of metric v at each occasion on its diagonal and the serial covariances of that metric off it. Two different metric vectors of the same unit covary as well, through the $T \times T$ matrix $\Sigma_g^{(vv')} := \mathrm{Cov}[Y_{ig}^{(v)}, Y_{ig}^{(v')}]$, whose diagonal holds the contemporaneous covariances between metrics v and v' and whose off-diagonal entries the lagged ones. Since Y_{ig} stacks the metric vectors, its covariance matrix assembles these blocks in the same order,

$$\Sigma_g = \mathrm{Var}[Y_{ig}] = \begin{pmatrix} \Sigma_g^{(11)} & \Sigma_g^{(12)} & \cdots & \Sigma_g^{(1V)} \\ \Sigma_g^{(12)\prime} & \Sigma_g^{(22)} & \cdots & \Sigma_g^{(2V)} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_g^{(1V)\prime} & \Sigma_g^{(2V)\prime} & \cdots & \Sigma_g^{(VV)} \end{pmatrix},$$

of dimension $VT \times VT$, which Equation 15 restricts only by requiring positive definiteness; in particular no separability between the metric and the occasion margin is assumed.

The group sample mean estimator stacks the cell means in the order of Y_{ig},

$$\bar{\mu}_g = (\bar{\mu}_g^{(1)\prime}, \ldots, \bar{\mu}_g^{(V)\prime})', \qquad \bar{\mu}_g^{(v)} = (\bar{\mu}_{gv1}, \ldots, \bar{\mu}_{gvT})', \qquad \bar{\mu}_{gvt} = \frac{1}{N_g} \sum_{i=1}^{N_g} Y_{igvt},$$

and the estimator of its sampling variance Σ_g / N_g becomes $\hat{\Sigma}_g / N_g$ with entries

$$\hat{\sigma}_{g,tt'}^{(vv')} := \frac{1}{N_g - 1} \sum_{i=1}^{N_g} (Y_{igvt} - \bar{\mu}_{gvt})(Y_{igv't'} - \bar{\mu}_{gv't'}).$$

Collecting the groups, the matrix Equation 17 is of dimension $GVT \times GVT$ and equals

$$\Sigma_{\sqrt{N}\bar{\mu}} = \begin{pmatrix} \frac{N}{N_1}\Sigma_1 & 0 & \cdots & 0 \\ 0 & \frac{N}{N_2}\Sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{N}{N_G}\Sigma_G \end{pmatrix},$$

the off-diagonal blocks vanishing because units in different groups are independent, and $\hat{\Sigma}_{\sqrt{N}\bar{\mu}}$ has the same form with $\hat{\Sigma}_g$ in place of Σ_g.

4 Applications

We now apply the general framework to several common tasks arising in multi-group randomized experiments: comparing two conditions, identifying optimal conditions, comparing multiple conditions against a baseline, testing equality of multiple conditions, and analyzing factorial designs. The key task in each case is to set up the parameter vector θ and the contrast matrix H. Everything else is supplied by the framework: a joint null $H\theta = 0$ is tested with the WTS Equation 5 whenever $\Sigma_{\hat{\vartheta}}$ is positive definite; row-wise decisions and simultaneous confidence intervals use the max-t procedure of Section 2.2; and a single row is analyzed as in Section 2.4 including power and sample size planning.

4.1 Contrasts

In the design of Example 1, $\theta = \mu = (\mu_1, \dots, \mu_G)'$ is estimated by the group sample means $\bar{\mu} = (\bar{\mu}_1, \dots, \bar{\mu}_G)'$ with the diagonal covariance estimator $\hat{\Sigma}_{\bar{\mu}} = \mathrm{diag}\{\hat{\sigma}_g^2/N_g\}$ of Section 3.1. For a contrast $c = (c_1, \dots, c_G)$ with $\sum_g c_g = 0$, the covariance terms of Equation 8 and Equation 9 vanish, since units in different groups are independent, and the contrast, its estimator, the variance and the variance estimator become

$$\begin{aligned} C = c\mu = \sum_{g=1}^{G} c_g \mu_g, \qquad \hat{C} = c\bar{\mu} = \sum_{g=1}^{G} c_g \bar{\mu}_g, \\ \mathrm{Var}[\hat{C}] = \sum_{g=1}^{G} \frac{c_g^2 \sigma_g^2}{N_g}, \qquad \widehat{\mathrm{Var}}[\hat{C}] = \sum_{g=1}^{G} \frac{c_g^2 \hat{\sigma}_g^2}{N_g}. \end{aligned} \tag{20}$$

The variance is exact (Equation 17), so only the normality of $\hat{C}$ and the consistency of the $\hat{\sigma}_g^2$ rest on large-sample arguments. With this standard error the tests and confidence sets of Section 2.4.2 apply by substitution, and the global test of Section 2.3 reduces to the squared statistic, since for $W = 1$ the matrix

$\hat{\Sigma}_{\hat{\vartheta}}$ in Equation 5 is the scalar $\widehat{\mathrm{Var}}[\hat{C}]$,

$$\begin{aligned} WTS &= (\hat{C} - C_0)\,\widehat{\mathrm{Var}}[\hat{C}]^{-1}(\hat{C} - C_0) \\ &= \frac{(\hat{C} - C_0)^2}{\sum_{g=1}^{G} c_g^2 \hat{\sigma}_g^2 / N_g} \\ &= Z^2 \xrightarrow[N\to\infty]{D} \chi_1^2, \end{aligned}$$

which rejects exactly when $|Z| > z_{1-\alpha/2}$ because $\chi^2_{1,1-\alpha} = z^2_{1-\alpha/2}$.

The pairwise comparison of groups g and g' is the contrast with $c_g = 1$, $c_{g'} = -1$ and all other weights zero, the average treatment effect $C = \mu_g - \mu_{g'} = \mathrm{ATE}(g, g')$ of Section 2.4.1, for which Equation 20 reduces to

$$\begin{aligned} \hat{C} &= \bar{\mu}_g - \bar{\mu}_{g'}, \qquad \widehat{\mathrm{Var}}[\hat{C}] = \frac{\hat{\sigma}_g^2}{N_g} + \frac{\hat{\sigma}_{g'}^2}{N_{g'}}, \\ Z &= \frac{\bar{\mu}_g - \bar{\mu}_{g'} - C_0}{\sqrt{\hat{\sigma}_g^2/N_g + \hat{\sigma}_{g'}^2/N_{g'}}}. \end{aligned} \tag{21}$$

This is the statistic of Welch's test for two means with unequal variances (Welch 1938, 1947; Jan and Shieh 2014); the two-group experiment is the special case $G = 2$.

The planning input of Section 2.4.3 becomes explicit as well, $\bar{V}_C = \sum_{g=1}^{G} c_g^2 \sigma_g^2 / p_g$, so that the sample size formula Equation 13 becomes

$$N \geq (z_{1-\alpha/2} + z_{1-\beta})^2 \delta^{-2} \sum_{g=1}^{G} c_g^2 \sigma_g^2 / p_g$$

with $N_g = p_g N$, and the noncentrality of Equation 6 is $\lambda_N = N\delta^2/\bar{V}_C = \lambda^2$. For a pairwise comparison these are the classical two-sample formulas (Equation 21).

The explicit form additionally allows optimizing the allocation. Minimizing $\bar{V}_C$ over $(p_1, \ldots, p_G)$ subject to $\sum_{g=1}^{G} p_g = 1$ yields the generalized *Neyman allocation* (Neyman 1934; Cochran 1977)

$$p_g^* = \frac{|c_g|\sigma_g}{\sum_{k=1}^{G} |c_k|\sigma_k}, \qquad \bar{V}_C^* = \Big(\sum_{g=1}^{G} |c_g|\sigma_g \Big)^2,$$

which is simultaneously optimal for power, minimum detectable effect and expected interval length, since all are monotone in $\bar{V}_C$ (List et al. 2011).

4.2 All-Pairwise Comparisons

When the goal is to identify the group with the largest parameter value θ_g, the natural family consists of *all pairwise comparisons*: $W = G(G-1)/2$ contrasts, each defined by a weight vector $c^{(w)}$ with a single $+1$, a single -1, and zeros elsewhere (Herberich et al. 2010). The resulting $W \times G$ contrast matrix is

$$H_{\text{pair}} = \begin{bmatrix} -1 & 1 & 0 & 0 & \dots & 0 & 0 \\ -1 & 0 & 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ -1 & 0 & 0 & 0 & \dots & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -1 & 1 \end{bmatrix}.$$

The family is analyzed row-wise with the max-t procedure of Section 2.2.

4.3 Many-to-One Comparisons

Comparing a single baseline condition against all others is referred to as a *many-to-one comparison* (Bretz et al. 2010). The corresponding $(G-1) \times G$ contrast matrix is

$$H = \begin{bmatrix} 1 & -1 & 0 & 0 & \dots & 0 \\ 1 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & 0 & \dots & -1 \end{bmatrix}.$$

All rows involve the baseline group, so the contrast estimators are correlated even when $\hat{\Sigma}_{\hat{\theta}}$ is diagonal, and the max-t procedure of Section 2.2 calibrates the family with this correlation.

4.4 Tests of Equality Across Groups

A fundamental use case is testing the global null hypothesis that all group parameters are equal,

$$H_0 : \theta_1 = \theta_2 = \cdots = \theta_G,$$

or, more generally, equality across a subset of $R < G$ parameters. The null is tested with the WTS Equation 5 using a full-row-rank matrix of successive differences; for all G parameters this is the $(G-1) \times G$ matrix

$$H_{\text{eq}} = \begin{bmatrix} 1 & -1 & 0 & \dots & 0 & 0 \\ 0 & 1 & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & -1 \end{bmatrix},$$

with $W = G - 1$ degrees of freedom ($R - 1$ for a subset). For group sample means in the design of Example 1, $\hat{\Sigma}_{\bar{\mu}}$ is diagonal and the WTS Equation 5 has the closed form

$$WTS = \sum_{g=1}^{G} \frac{N_g}{\hat{\sigma}_g^2} (\bar{\mu}_g - \tilde{\mu})^2, \qquad \tilde{\mu} := \frac{\sum_{g=1}^{G} (N_g/\hat{\sigma}_g^2)\, \bar{\mu}_g}{\sum_{g=1}^{G} N_g/\hat{\sigma}_g^2}, \tag{22}$$

the precision-weighted sum of squared deviations of the group means from their precision-weighted average. It does not depend on the choice of the difference matrix, since replacing H by AH with A invertible leaves Equation 5 unchanged, and is the large-sample form of the Welch–James statistic for the heteroscedastic one-way layout (James 1951; Welch 1951; Algina and Olejnik 1984).

Under local alternatives the WTS is noncentral $\chi^2_{G-1}(\lambda_N)$ (Equation 6), with the noncentrality of the same form in population quantities,

$$\lambda_N = \sum_{g=1}^{G} \frac{N_g}{\sigma_g^2} (\mu_g - \tilde{\mu})^2, \tag{23}$$

where $\tilde{\mu}$ now averages the μ_g with weights N_g/σ_g^2; power and sample size follow from Equation 7. Under balanced allocation and homoscedasticity, $N_g = N/G$ and $\sigma_g = \sigma$, this is $\lambda_N = Nf^2$ with Cohen's $f^2 = \sum_g (\mu_g - \mu_\cdot)^2/(G\sigma^2)$ and $\mu_\cdot := G^{-1} \sum_g \mu_g$, the noncentrality of classical ANOVA power analysis (Cohen 1988).

The classical one-way ANOVA F test is the homoscedastic special case. Its statistic

$$F = \frac{\sum_{g=1}^{G} N_g (\bar{\mu}_g - \bar{\mu}_\cdot)^2/(G-1)}{\hat{\sigma}^2_{\text{pool}}}, \qquad \hat{\sigma}^2_{\text{pool}} := \frac{1}{N-G} \sum_{g=1}^{G} (N_g - 1)\, \hat{\sigma}_g^2, \quad \bar{\mu}_\cdot := \frac{1}{N} \sum_{g=1}^{G} N_g \bar{\mu}_g, \tag{24}$$

satisfies $(G-1)F = WTS$ with $\hat{\sigma}^2_{\text{pool}}$ in place of every $\hat{\sigma}_g^2$ in Equation 22, since the weights are then proportional to N_g and $\tilde{\mu}$ becomes the grand mean $\bar{\mu}_\cdot$. Under H_0 and homoscedasticity $(G-1)F \xrightarrow[N\to\infty]{D} \chi^2_{G-1}$, so the two tests are asymptotically equivalent.

4.5 Factorial Designs

Factorial designs study the effects of multiple experimental factors simultaneously. A *general factorial design* with K factors $f_1, \dots, f_K$, each having L_{f_k} levels, yields $G = \prod_{k=1}^{K} L_{f_k}$ treatment groups with mean vector $\mu = (\mu_1, \dots, \mu_G)'$, a 2^K *design* when every factor has two levels; either all groups are run, or a fraction of them (Section 4.5.2). The objective is to estimate and test *main effects* (the marginal effect of each factor) and *interaction effects* (the joint effect of two or more

factors). For a nonempty subset $\mathcal{S} \subseteq \{1, \dots, K\}$ of factors, the null hypothesis of no effect is

$$H_0(\mathcal{S}) : H_{\mathcal{S}}\mu = 0,$$

where $\mathcal{S} = \{k\}$ gives the main effect of f_k (written H_k) and $\mathcal{S} = \{j, k\}$ the two-factor interaction of f_j and f_k (written H_{jk}). Each $H_{\mathcal{S}}$ enters the WTS Equation 5 for a joint test, single rows are analyzed as in Section 2.4, and several rows simultaneously as in Section 2.2, without assuming normality or homoscedasticity.

4.5.1 Full Factorial Designs

In a full factorial design all G treatment combinations are run. Order the groups lexicographically by their factor levels with the last factor varying fastest, so that μ lists the cell means in this order. For each factor f_k choose a *within-factor contrast matrix* $\Gamma_k \in \mathbb{R}^{(L_{f_k}-1)\times L_{f_k}}$ with full row rank and zero row sums, for instance successive differences as in Section 4.4, and let $m_k' := L_{f_k}^{-1}\mathbb{1}_{L_{f_k}}'$ denote the averaging row. The contrast matrix for $\mathcal{S}$ is the Kronecker product

$$H_{\mathcal{S}} = \bigotimes_{k=1}^{K} A_k, \qquad A_k = \begin{cases} \Gamma_k, & k \in \mathcal{S}, \\ m_k', & k \notin \mathcal{S}, \end{cases} \tag{25}$$

which compares the levels of the factors in $\mathcal{S}$ and averages over the levels of all remaining factors (Pauly et al. 2015; see also Algina and Olejnik 1984). It has $W_{\mathcal{S}} = \prod_{k\in\mathcal{S}}(L_{f_k} - 1)$ rows, each summing to zero, and full row rank, so Equation 5 applies with $W = W_{\mathcal{S}}$ degrees of freedom. All admissible Γ_k span the orthogonal complement of $\mathbb{1}_{L_{f_k}}$ and therefore differ only by invertible factors, which leave Equation 5 unchanged, so the WTS does not depend on the coding; single rows do, and should match the comparisons of substantive interest.

Example 5 (3×2 full factorial). *Factor A has three levels and factor B has two levels, for instance three discount levels crossed with two shipping options, and all $G = 6$ treatment combinations are run.*

Let μ_{ab} denote the mean of the cell with $A = a$ and $B = b$, so that

$$\mu = (\mu_{11}, \mu_{12}, \mu_{21}, \mu_{22}, \mu_{31}, \mu_{32})'.$$

With successive differences,

$$\Gamma_A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \qquad \Gamma_B = (-1 \;\; 1),$$

$$m_A' = \left(\tfrac{1}{3} \;\; \tfrac{1}{3} \;\; \tfrac{1}{3}\right), \qquad m_B' = \left(\tfrac{1}{2} \;\; \tfrac{1}{2}\right),$$

and Equation 25 yields

$$H_A = \Gamma_A \otimes m'_B = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix},$$

$$H_B = m'_A \otimes \Gamma_B = \tfrac{1}{3}\begin{bmatrix} -1 & 1 & -1 & 1 & -1 & 1 \end{bmatrix},$$

$$H_{AB} = \Gamma_A \otimes \Gamma_B = \begin{bmatrix} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}.$$

Write the marginal means and the effect of B at level a of A as

$$\begin{aligned} \mu_{a\cdot} &:= \tfrac{1}{2}(\mu_{a1} + \mu_{a2}), \\ \mu_{\cdot b} &:= \tfrac{1}{3}(\mu_{1b} + \mu_{2b} + \mu_{3b}), \\ \delta_B(a) &:= \mu_{a2} - \mu_{a1}. \end{aligned}$$

The contrasts are

$$H_A\mu = \begin{bmatrix} \mu_{2\cdot} - \mu_{1\cdot} \\ \mu_{3\cdot} - \mu_{2\cdot} \end{bmatrix},$$

$$H_B\mu = \mu_{\cdot 2} - \mu_{\cdot 1},$$

$$H_{AB}\mu = \begin{bmatrix} \delta_B(2) - \delta_B(1) \\ \delta_B(3) - \delta_B(2) \end{bmatrix},$$

with $W = 2$, 1, and 2 degrees of freedom. The hypothesis $H_{AB}\mu = 0$ states that the effect of B is the same at all three levels of A; each of its rows is the 2×2 interaction contrast of Section 2.4.1 applied to two adjacent levels of A.

4.5.2 Fractional Factorial Designs

With many factors or levels a full factorial design becomes impractical: the number of groups grows multiplicatively, and every additional group reduces the group sizes N_g on which the large-sample approximations rest. A *fractional factorial design* runs only a suitably chosen subset of the treatment combinations, which then form the $G < \prod_k L_{f_k}$ groups of the analysis. Effects that the design cannot separate are *aliased*, and an effect is identified only under an assumption on its aliases, conventionally that they are zero (Box et al. 1978; Wu and Hamada 2011; Mukerjee and Wu 2006). A design is specified by its design table, which lists the coded factor levels of each group, $-$ and $+$ or -1 and $+1$ for two-level factors; the following example derives the contrasts from this table.

Example 6 (2^{5-2} fractional factorial). *Five two-level factors A, B, C, D, E are studied in $G = 2^{5-2} = 8$ groups, a quarter of the $2^5 = 32$ treatment combinations.*

Table 1 is the design table: a full 2^3 design in the factors A, B, C, with the levels of the remaining two factors set by the *generators* $D = AB$ and $E = AC$, where a product of factor letters denotes the elementwise product of their coded columns (the *product column* of that effect).

Table 1: Design table of the 2^{5-2} fraction with generators $D = AB$ and $E = AC$.

group g	A	B	C	$D = AB$	$E = AC$
1	$-$	$-$	$-$	$+$	$+$
2	$-$	$-$	$+$	$+$	$-$
3	$-$	$+$	$-$	$-$	$+$
4	$-$	$+$	$+$	$-$	$-$
5	$+$	$-$	$-$	$-$	$-$
6	$+$	$-$	$+$	$-$	$+$
7	$+$	$+$	$-$	$+$	$-$
8	$+$	$+$	$+$	$+$	$+$

Here two effects are aliased exactly when their product columns coincide on the eight groups. Of the $2^5 - 1 = 31$ effects, 28 fall into seven alias classes of four; the product columns of the remaining three, ABD, ACE, and $BCDE$, equal $+1$ in every group and yield no contrast, and the shortest of them has three letters, so the design has resolution III (Box and Hunter 1961, 319). Each alias class gives one row of the contrast matrix H: the columns $A, \dots, E$ of Table 1, read from top to bottom, and, for the two classes without a main effect, the product columns BC and BE. Scaled by $\frac{1}{4}$, each row gives the average of the four group means with sign $+$ minus the average of the four with sign $-$:

$$H = \frac{1}{4}\begin{bmatrix} -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1 \end{bmatrix},$$

$$H\mu = \begin{bmatrix} \tau_A + \tau_{BD} + \tau_{CE} + \tau_{ABCDE} \\ \tau_B + \tau_{AD} + \tau_{CDE} + \tau_{ABCE} \\ \tau_C + \tau_{AE} + \tau_{BDE} + \tau_{ABCD} \\ \tau_D + \tau_{AB} + \tau_{BCE} + \tau_{ACDE} \\ \tau_E + \tau_{AC} + \tau_{BCD} + \tau_{ABDE} \\ \tau_{BC} + \tau_{DE} + \tau_{ABE} + \tau_{ACD} \\ \tau_{BE} + \tau_{CD} + \tau_{ABC} + \tau_{ADE} \end{bmatrix}, \tag{26}$$

where $\mu = (\mu_1, \dots, \mu_8)'$ collects the means of the eight groups and $\tau_{\mathcal{S}}$ is the effect of the factor subset $\mathcal{S}$ in the full 2^5 design,

$$\tau_{\mathcal{S}} := \frac{1}{16} \sum_{x \in \{-1,+1\}^5} \Big(\prod_{k \in \mathcal{S}} x_k \Big) \mu(x),$$

with $\mu(x)$ the mean of the treatment combination with coded levels $x = (x_A, \dots, x_E)$, so that $\mu_g = \mu(x)$ for row g of Table 1; $\tau_{\mathcal{S}}$ equals $2^{-(|\mathcal{S}|-1)}$ times

the contrast Equation 25 with $\Gamma_k = (-1 \;\; 1)$. The matrix H has full row rank $W = 7 = G - 1$, so Equation 5 applies to any selection of its rows. The first row of $H\mu$ is the main effect τ_A if the interactions BD, CE, and $ABCDE$ are zero, and analogously for the other rows; if all interactions are zero, the first five rows are the five main effects, while the last two rows contain interactions only and provide a partial check of that assumption.

Appendix

Derivations for the planning formulas of Section 2.4.3, on which Section 2.3 draws for $W = 1$.

Derivation of Equation 11. Decomposing the test statistic around the truth gives

$$Z = \frac{\hat{C} - C_0}{\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}} = \frac{\hat{C} - C}{\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}} + \frac{\delta}{\sqrt{\widehat{\mathrm{Var}}[\hat{C}]}},$$

where the first term is asymptotically standard normal — by the same argument that gives the null distribution of Z — and the second equals λ times a factor that tends to one in probability, since $N\,\widehat{\mathrm{Var}}[\hat{C}] \xrightarrow{P} \bar{V}_C$. Setting $\delta = 0$ recovers the null distribution, so one formula covers both hypotheses. The approximation is rigorous under local alternatives $\delta = \delta^*/\sqrt{N}$ along which the two conditions of Section 2.1 continue to hold: then $\lambda = \delta^*/\sqrt{\bar{V}_C}$ and $Z - \lambda \xrightarrow{D} N(0,1)$ (Vaart 1998). For fixed $\delta \neq 0$ the noncentrality diverges and power tends to one, but $Z - \lambda$ is in general not asymptotically standard normal, because the estimation error of the variance is multiplied by the diverging λ. For a $\sqrt{N}$-consistent variance estimator the discrepancy is of order $\delta/\sqrt{\bar{V}_C} = \lambda/\sqrt{N}$, so Equation 11 is accurate whenever λ is small relative to $\sqrt{N}$, that is, at the effects for which power is not already near one.

Derivation of Equation 12. Under the normal approximation Equation 11, the power of the two-sided test is

$$\begin{aligned}\pi &= P\big[|Z| > z_{1-\alpha/2}\big] \\ &= \Phi_{0,1}(\lambda - z_{1-\alpha/2}) + \Phi_{0,1}(-\lambda - z_{1-\alpha/2});\end{aligned}$$

at $\delta = 0$ the right side equals α, confirming the size of the test. The second term is the probability of rejecting with the wrong sign. It is negligible unless λ is close to zero, and omitting it understates power, so Equation 12 is conservative for planning.

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